

Learning to unlearn: The role of negative evidence in morphophonological learning

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Morphophonological learning

What is morphophonology?

Morphophonology: The study of how phonological forms are used to express grammatical/morphological functions

- ▶ Learning morphophonology is quite challenging

Morphophonological learning

Why is morphophonology interesting?

Challenging, because a form has multiple meanings:

- ▶ In German, [ə] marks PLURAL, but also SUBJECT-VERB AGREEMENT, ADJECTIVE AGREEMENT or no function at all
- ▶ PLURAL → *Wale* [va:lə] “whales”
- ▶ SUBJECT-VERB AGREEMENT → *(Ich) denke* [dɛŋkə] “(I) think”
- ▶ ADJECTIVE AGREEMENT → *Tolle (Jacke)* [tɔlə] “Great (jacket)”
- ▶ ∅ → *Jacke* [jakə] “jacket”

Morphophonological learning

Why is morphophonology interesting?

Challenging, because multiple forms have the same meaning:

- ▶ PLURAL can be expressed through different forms ([ə] [s])
- ▶ *Balkone/Balkons* [balko:nə]/[balko:n^s] “balconies” →
PLURAL

Forms additionally vary in frequency and salience!

Learning theories

Learning theories

Statistical learning

Only positive evidence has an effect on learning¹

Error-driven learning

Both positive **and** negative evidence (cue is present, outcome is absent) have an effect on learning²

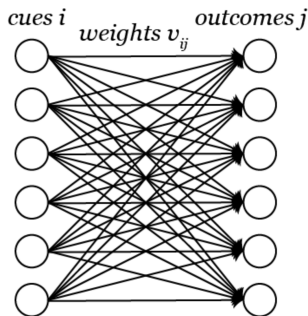
¹(Aslin, Saffran, & Newport, 1998; Maye, Werker, & Gerken, 2002; Saffran, 2020; Saffran, Aslin, & Newport, 1996; Saffran & Kirkham, 2018)

²Harmon, Idemaru, & Kapatsinski (2019), Nixon (2020), Nixon & Tomaschek (2021), Olejarczuk, Kapatsinski, & Baayen (2018), Ramscar, Dye, & McCauley (2013), and Ramscar et al. (2010)

Learning theories

Rescorla-Wagner learning rule

Error-driven learning can be simulated in a fully connected two-layer (end-to-end) network using the *Rescorla-Wagner* equation (Rescorla & Wagner, 1972)



Activation weights between *Cues* and *Outcomes* (Hoppe et al., 2022)

Learning theories

Rescorla-Wagner learning rule

Updating the weight matrix

$$w_{ij}^{t+1} = w_{ij}^t + \Delta w_{ij}^t \quad (1)$$

Rescorla-Wagner equation

$$\Delta w_{ij}^t = \begin{cases} \text{(a) } 0 & \text{if } \textit{Absent}(c_i, t) \\ \text{(b) } \alpha_i \beta_1 \left(\lambda_j - \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Present}(o_j, t) \\ \text{(c) } \alpha_i \beta_2 \left(0 - \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Absent}(o_j, t) \end{cases} \quad (2)$$

Learning theories

Research question

Does unlearning from negative evidence benefit morphophonological learning?

- ▶ if a cue occurs without outcome = association strength is decreased (negative evidence)
- ▶ For comprehension: *cues* (= phonological word-forms) predict *outcomes* (= semantics/morphological category)
- ▶ This was tested in the artificial language learning paradigm (Culbertson, 2012)

The artificial language

Items

German adult speakers were taught a novel morphological pattern:

Singular (SG)	<a>	CVC[a]C
Plural (PL)	<i>	CVCC[i]
Diminutive (DM)	<ić>	CVCC[i ^h tʃ]
Diminutive-Plural (DMPL)	<ić> + <i>	CVCC[i ^h tʃ][i]

Morphological pattern

Phonological word forms served as cues for the morphological outcomes

The artificial language

Items



[be.fan]



[bef.ni]



[bef.ni.tʃ]



[bef.ni.tʃi]

Word forms and their grammatical function

The artificial language

Conditions and groups

Group	Learning I	Learning II	Test cues
DM_absent condition			
DM: No_Unlearning	X	X	X
DM: Unlearning	X	X	X
PL_absent condition			
PL: No_Unlearning	X	X	X
PL: Unlearning	X	X	X

Overview of the four different groups

The groups within the conditions received the same input: Only learning block order differed!

The artificial language

DM_absent: No_Unlearning

Group	Learning I	Learning II	Test cues
DM: No_Unlearning	$[i] \rightarrow \text{PL}$	$[\widehat{it}i] \rightarrow \text{DMPL}$	$[\widehat{it}i] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$



(1) [bef.ni]

(2) [bef.ni.t $\widehat{f}$ i]



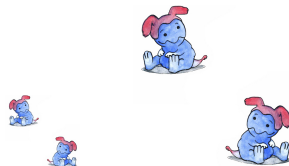
(1+2) [be.fan]

[bef.nit $\widehat{f}$ i]

The artificial language

DM_absent: Unlearning

Group	Learning I	Learning II	Test cues
DM: Unlearning	$[\widehat{it}i] \rightarrow \text{DMPL}$	$[i] \rightarrow \text{PL}$	$[\widehat{it}f] \rightarrow \text{DM}; [i] \rightarrow \text{PL}$



(1) $[\text{bef.ni.t}\widehat{i}]$

(2) $[\text{bef.ni}]$



(1+2) $[\text{be.fan}]$

$[\text{bef.nit}\widehat{f}]$

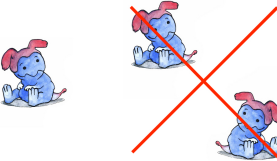
The artificial language

PL_absent: No_Unlearning

Group	Learning I	Learning II	Test cues
PL: No_Unlearning	$[\widehat{it}\widehat{f}] \rightarrow \text{DM}$	$[\widehat{it}\widehat{ni}] \rightarrow \text{DMPL}$	$[\widehat{it}\widehat{f}] \rightarrow \text{DM}; [\widehat{ni}] \rightarrow \text{PL}$



 (1) $[\text{bef.nit}\widehat{f}]$ (2) $[\text{bef.ni.t}\widehat{f}\widehat{ni}]$



 (1+2) $[\text{be.fan}]$ $[\text{bef.ni}]$

The artificial language

PL_absent: Unlearning

Group	Learning I	Learning II	Test cues
PL: Unlearning	$[\widehat{it}\widehat{i}] \rightarrow \text{DMPL}$	$[\widehat{it}\widehat{j}] \rightarrow \text{DM}$	$[\widehat{it}\widehat{j}] \rightarrow \text{DM}; [\widehat{i}] \rightarrow \text{PL}$



(1) $[\text{bef.ni.t}\widehat{j}\widehat{i}]$

(2) $[\text{bef.nit}\widehat{j}\widehat{j}]$



(1+2) $[\text{be.fan}]$

$[\text{bef.ni}]$

Learning simulation

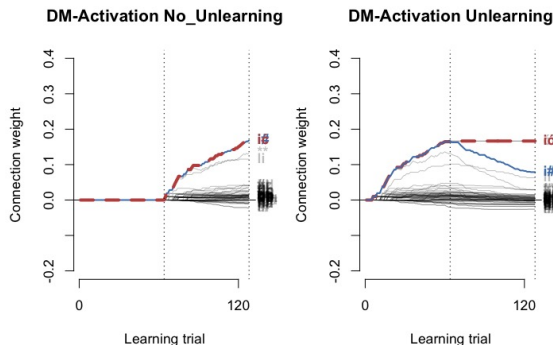
Rescorla-Wagner learning rule

Computational simulations for the four learning groups using the `edl`-package (van Rij & Hoppe, 2021)

- ▶ *Cues*: Bigrams of the word forms (computed with the `ndl`-package (Arppe et al., 2018))
 - ▶ e.g. the pseudoword `goplići` was split into its bigrams
 - ▶ `#g`, `go`, `op`, `pl`, `li`, `ić`, `ći`, `i#`
- ▶ *Outcomes*: Morphological category (SG, PL, DM)

DM_absent

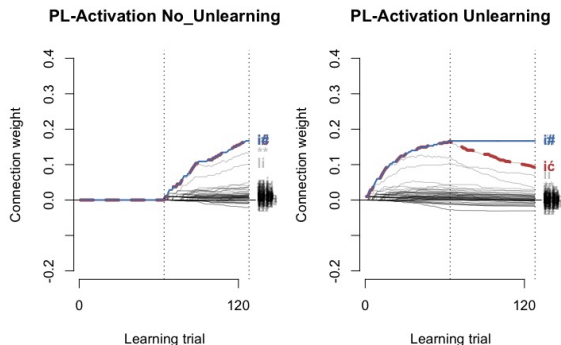
Results No_Unlearning (left) vs. Unlearning (right)



Association between i [i] and $i\hat{c}$ [$i\hat{f}$] to PL due to DMPL exposure

PL_absent

Results Unlearning (left) vs. No_Unlearning (right)



Association between i [i] and $i\hat{c}$ [iĉ] to DM due to DMPL exposure

Learning simulation

Differences between learning orders

No_Unlearning	(1) $A \rightarrow X$	(2) $AB \rightarrow XY$	(3) $A \rightarrow Y, B \rightarrow Y$
Unlearning	(1) $AB \rightarrow XY$	(2) $A \rightarrow X$	(3) $A \rightarrow Y, B \rightarrow Y$

Schematic differences between learning order

- No_Unlearning groups are expected to learn the withhold category worse, as the competing, non-discriminative cue receives the same activation weight

Learning experiment

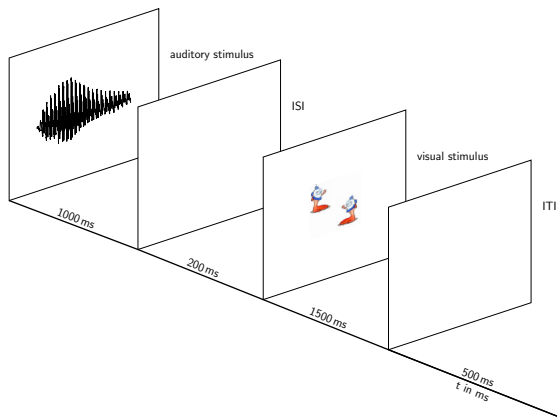
Participants

116 participants tested (DM_absent: 32 in No_Unlearning, 25 in Unlearning; PL_absent: 30 in No_Unlearning and 29 in Unlearning)

- ▶ Experiments programmed in *PsychoPy* (Peirce et al., 2019) and run online through *Pavlovia* (Bridges et al., 2020)
- ▶ Only the word forms for PL and DM morphological categories were tested
- ▶ Half of the forms were novel, half were repeated from the learning blocks

Learning experiment

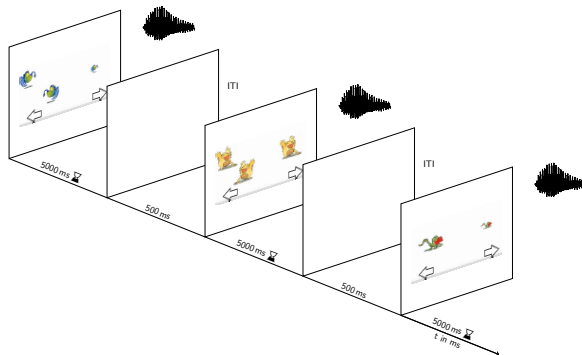
Procedure



Time course of a single trial for the learning blocks I and II (128 in total).

Learning experiment

Procedure



Time course of the test phase (three trials are shown). 64 trials (32 PL, 32 DM)

Predictions

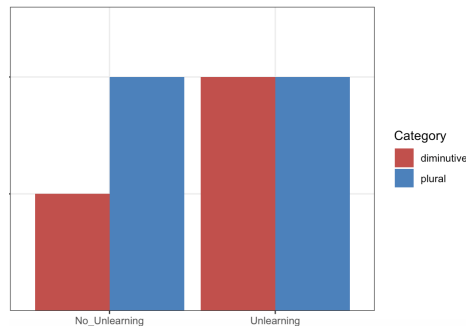
DM_absent condition

$[i] \rightarrow \text{Plural}$

$\text{No_Unlearning} = \text{Unlearning}$

$[itf] \rightarrow \text{Diminutive}$

$\text{No_Unlearning} < \text{Unlearning}$



Predictions

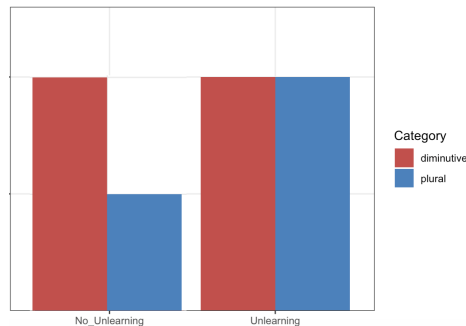
PL_absent condition

$[i] \rightarrow \text{Plural}$

No_Unlearning < Unlearning

$[itf] \rightarrow \text{Diminutive}$

No_Unlearning = Unlearning



Learning experiment

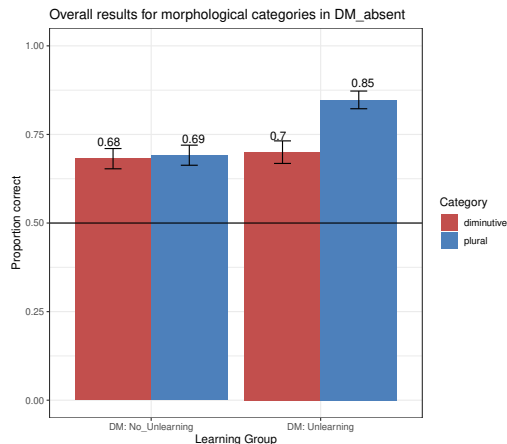
Results

Overall 7424 data points (116 participants $\times$ 64 trials) analyzed in R (R Core Team, 2021)

- ▶ *Generalized linear mixed regression model*-analysis using `lmerTest` (Kuznetsova, Brockhoff, & Christensen, 2017)
- ▶ Random intercepts for Word and Participant
- ▶ Dependent variable: accuracy – 0 (wrong), 1 (right)

Learning experiment

Results



Probability of correct choice for diminutives and plurals in DM_absent. Error bars show the 95% confidence interval.

Learning experiment

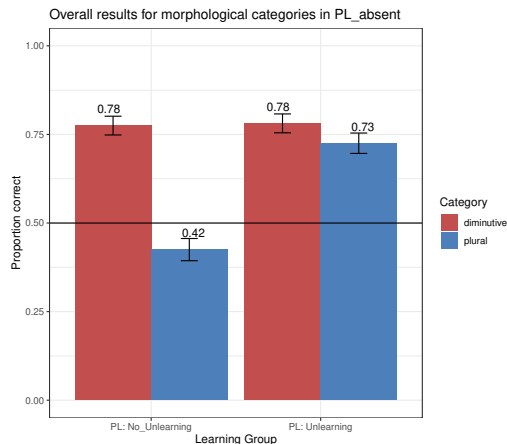
Results

Summary of the GLMM for accuracy during the test phase for DM_absent (No_Unlearning and Unlearning.)

Fixed effects					
	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept)	2.3668	0.5385	4.3950	<0.0001	***
Group = Unlearning	0.2692	0.3728	0.7220	0.4702	
Category = Plural	0.0631	0.1458	0.4330	0.6653	
Sequence	0.0145	0.0024	5.9460	<0.0001	***
Age	-0.0643	0.0173	-3.7170	0.0002	***
Group: Category	0.9585	0.1871	5.1230	<0.0001	***

Learning experiment

Results



Probability of correct choice for diminutives and plurals in PL_absent. Error bars show the 95% confidence interval.

Learning experiment

Results

Summary of the GLMM for accuracy during the test phase for PL_absent (No_Unlearning and Unlearning).

Fixed effects					
	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept)	1.8804	0.3689	5.097	<0.0001	***
Group = Unlearning	-0.1091	0.5092	-0.2140	0.8304	
Category = Plural	-2.4821	0.1643	-15.103	<0.0001	***
Sequence	0.0064	0.0024	2.6280	0.0086	**
Group: Category	2.0628	0.1916	10.7680	<0.0001	***

Learning experiment

Did *Unlearning* have an effect?

- ▶ Withhold DM-category in DM_absent **did not benefit** from unlearning of the non-discriminative cue
 - ▶ But: Participants performed better in the PL-category instead (85% accuracy vs. 69%)
- ▶ Withhold PL-category in PL_absent **did benefit** from unlearning of the non-discriminative cue
 - ▶ Better performance for the PL-category (73% accuracy vs. 42%)
- ▶ Learning order and Unlearning affected learning

Conclusion

- ▶ Behavioral experiment confirmed learning predictions obtained from ed1-simulations
- ▶ Unlearning (and learning order differences) can benefit learning
- ▶ Unlearning through negative evidence occurs even in morphophonological learning
- ▶ But: only if the non-discriminative, more salient *cue* is unlearned
- ▶ $[\widehat{itf}]$ more salient than $[i]$, because $[i]$ was overlapping in DM and PL
- ▶ while $[\widehat{tj}]$ was unique $\rightarrow$ more attended to

Conclusion

- ▶ Less salient $[i]$ cue was overshadowed by the more salient $[\widehat{itf}]$ cue
- ▶ This explains why even without unlearning $[\widehat{itf}] \rightarrow \text{DM}$ was still learned in the DM_absent condition
- ▶ Both positive and negative evidence matter during morphophonological learning

Question & Discussion



- Are there any questions? Comments?

References I



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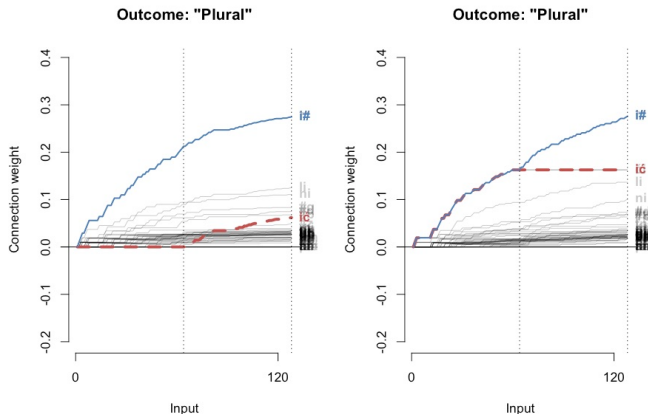
Question & Discussion



- Are there any questions? Comments?

Appendix

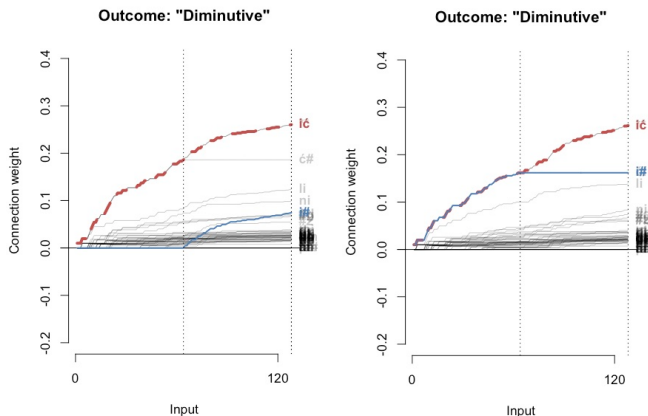
Simulation I: Plural activation (DM_absent)



DM_absent: No_Unlearning (left) vs. Unlearning (right)

Appendix

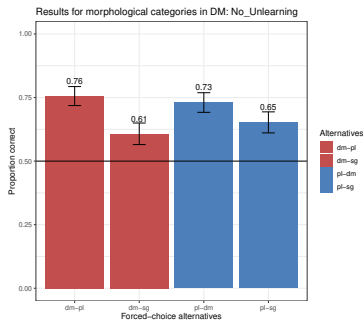
Simulation II: Diminutive activation (PL_absent)



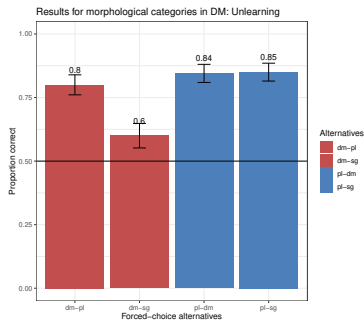
PL_absent: No_Unlearning (left) vs. Unlearning (right)

Appendix

Results I: 2AFC-alternatives



Proportions of correct choice for
DM: No_Unlearning

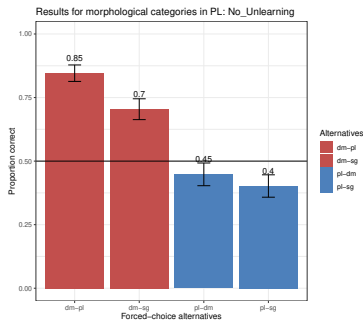


Proportions of correct choice for
DM: Unlearning

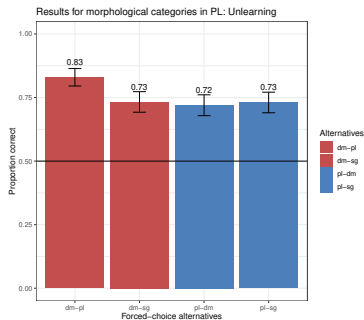
Proportions of correct choice for forced-choice contrasts in DM_absent

Appendix

Results I: 2AFC-alternatives



Proportions of correct choice for
PL: No_Unlearning

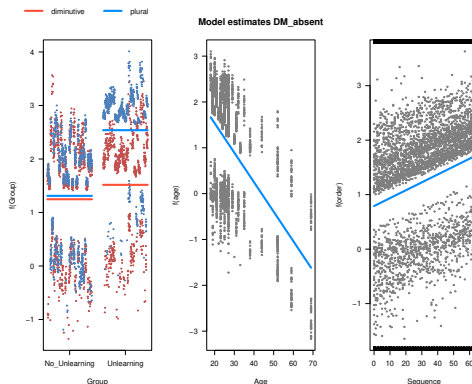


Proportions of correct choice for
PL: Unlearning

Proportions of correct choice for forced-choice contrasts in PL_absent

Appendix

Results II: GLMM results

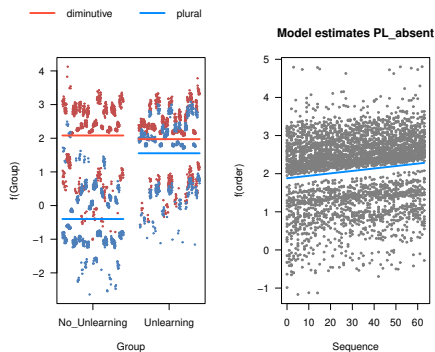


Model estimates for the results of the
DM_absent-groups (DM: No_Unlearning, DM:
Unlearning)

GLMM estimates for DM_absent

Appendix

Results II: GLMM results



Model estimates for the results of the PL_absent-groups (PL: No_Unlearning, PL: Unlearning)

GLMM estimates for PL_absent